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Saveetha Institute of Medical And Technical Sciences
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Image Acquisition Process

Subtitle: Challenges, CCD vs CMOS Sensors, and System Model

ITA0515-COMPUTER VISION

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INTRODUCTION

- Image acquisition is the process of capturing images using a camera or sensor.
- It converts a real-world scene into a digital image.
- It is the first step in image processing.
- Used in medical imaging, surveillance, robotics, and smartphones.
- **Image:** Camera capturing an image.

OBJECTIVES

- Understand the image acquisition process.
- Identify real-world image capturing challenges.
- Learn the functions of image acquisition components.
- Compare CCD and CMOS sensors.
- Study the image acquisition system model.

IMAGE ACQUISITION PROCESS

Flow Diagram:

Object

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Camera Lens

↓

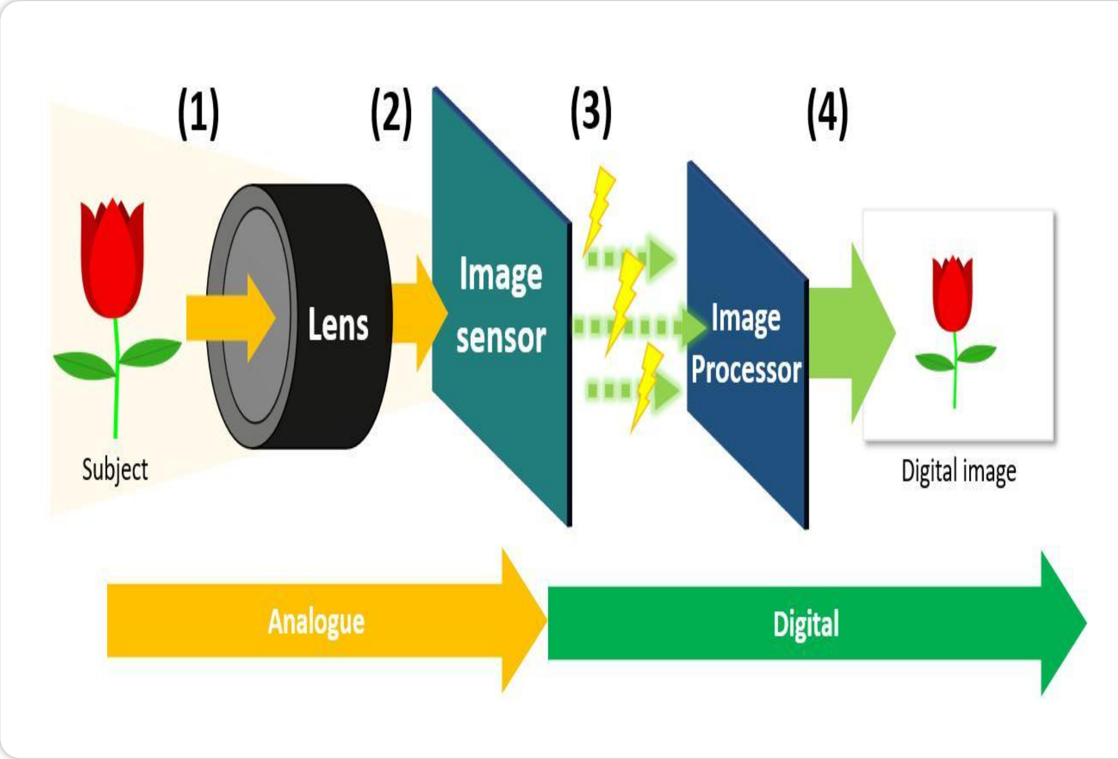
CCD / CMOS Sensor

↓

ADC

↓

Digital Image



Real-World Challenges

Challenges in Image Acquisition

- Low lighting
- Motion blur
- Image noise
- Camera shake
- Weather conditions

Effects:

- Poor image quality
- Loss of details
- Incorrect detection

- Overexposure and underexposure

CCD vs CMOS Sensors

CCD	CMOS
Better image quality	Faster processing
Low noise	Low power consumption
High power usage	Low cost
Expensive	Widely used in smartphones

Conclusion:

- CCD ⟶ Better quality
- CMOS ⟶ Faster and energy efficient

CONCLUSION

- ❖ Image acquisition converts real scenes into digital images.
- ❖ CCD and CMOS are the main image sensors.
- ❖ Image quality depends on lighting, motion, and sensor performance.
- ❖ CMOS sensors are commonly used because they are fast and affordable.

Thank you